

DSA 8020 R Lab 8: Randomized Complete Block Designs and Factorial Designs

saransh rakshak

Contents

A researcher wishes to determine whether four different tips produce different hardness readings on a Rockwell hardness tester, which operates by pressing a tip into a metal test ‘coupon.’ Since the coupons are large enough to test four tips, a RCBD (Randomized Complete Block Design) can be used, with one coupon serving as a block. The dataset ‘tip_hardness.csv’ can be found on Canvas.

1. Load the data into R.

Code:

```
tip_data <- read.csv("tip_hardness.csv")
head(tip_data)
```

```
##   Obs Tip Hardness Coupon
## 1   1   1     9.3      1
## 2   2   1     9.4      2
## 3   3   1     9.6      3
## 4   4   1    10.0      4
## 5   5   2     9.4      1
## 6   6   2     9.3      2
```

```
summary(tip_data)
```

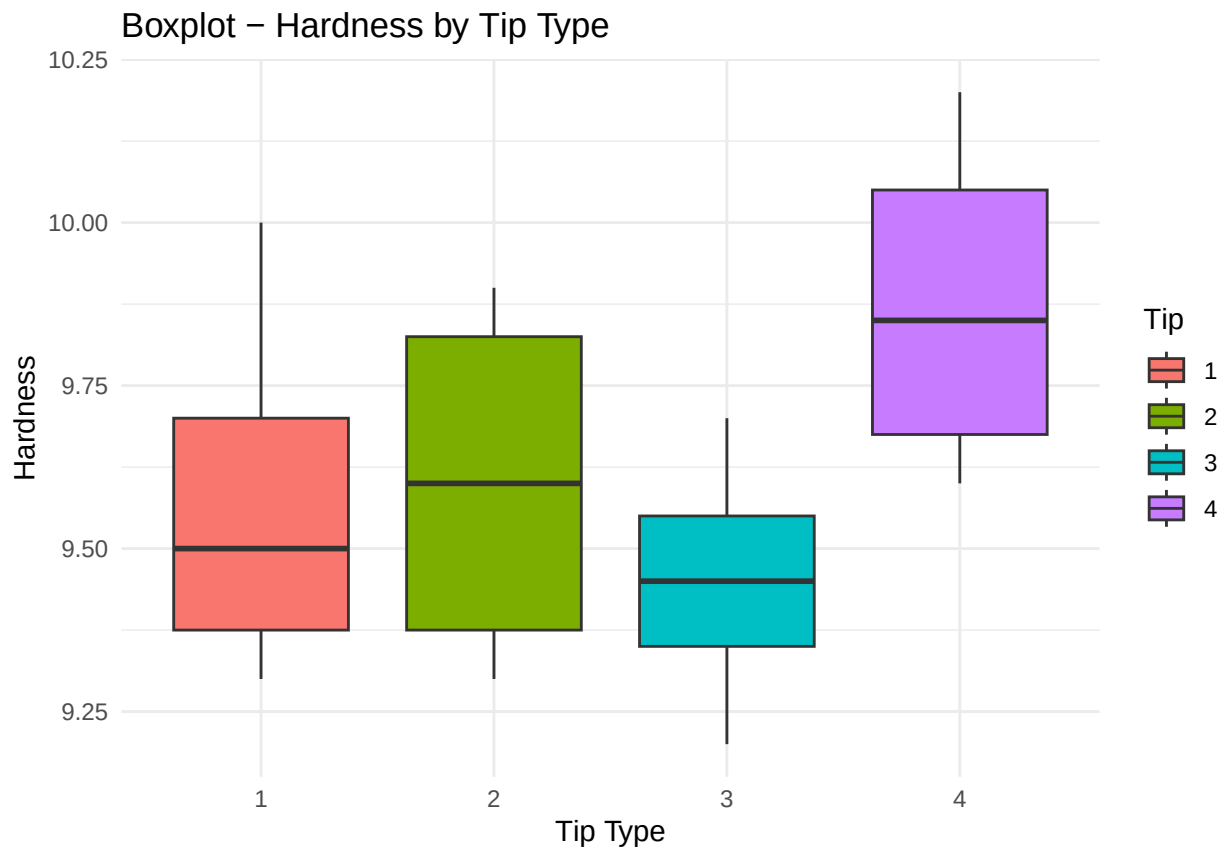
```
##      Obs      Tip      Hardness      Coupon
## Min.   : 1.00  Min.   :1.00  Min.    : 9.200  Min.    :1.00
## 1st Qu.: 4.75  1st Qu.:1.75  1st Qu.: 9.400  1st Qu.:1.75
## Median : 8.50  Median :2.50  Median : 9.600  Median :2.50
## Mean   : 8.50  Mean   :2.50  Mean   : 9.625  Mean   :2.50
## 3rd Qu.:12.25  3rd Qu.:3.25  3rd Qu.: 9.825  3rd Qu.:3.25
## Max.   :16.00  Max.   :4.00  Max.    :10.200  Max.    :4.00
```

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2. Make side-by-side boxplots by treatment.

Code:

```
# converting tip to categorical
tip_data <- tip_data %>%
  mutate(Tip = as.factor(Tip))

ggplot(tip_data, aes(x = Tip, y = Hardness, fill = Tip)) +
  geom_boxplot() +
  labs(title = "Boxplot - Hardness by Tip Type", x = "Tip Type", y = "Hardness") +
  theme_minimal()
```



3. Write down the effects model and explain each term within the model, including the assumptions concerning the random error.

Answer:

RCBD Effects Model: $Y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}$

- Y_{ij} : Hardness value for i th Tip in j th Coupon.
- μ : Mean across all Tips/Coupons.
- τ_i : Effect of i th Tip (treatment).
- β_j : Effect of j th Coupon (block).
- ϵ_{ij} : Random error. Assumptions are:

- Independent: ϵ_{ij} must be independent for each observation.
- Normality: ϵ_{ij} must follow normal distribution.
- Homoscedasticity: ϵ_{ij} must have constant variance across treatment/blocks.

4. Perform an overall F -test to the treatment effects using ANOVA, state the hypotheses, p -value, and conclusion.

- Null (H_0): There is no difference in hardness between tip types.
- Alternate (H_A): At least one tip creates a different hardness.

Code:

```
tip_anova <- aov(Hardness ~ Tip + Coupon, data = tip_data)
summary(tip_anova)
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Tip         3  0.3850   0.1283    9.769  0.00196 **
## Coupon      1  0.7605   0.7605   57.893 1.05e-05 ***
## Residuals  11  0.1445   0.0131
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Answer:

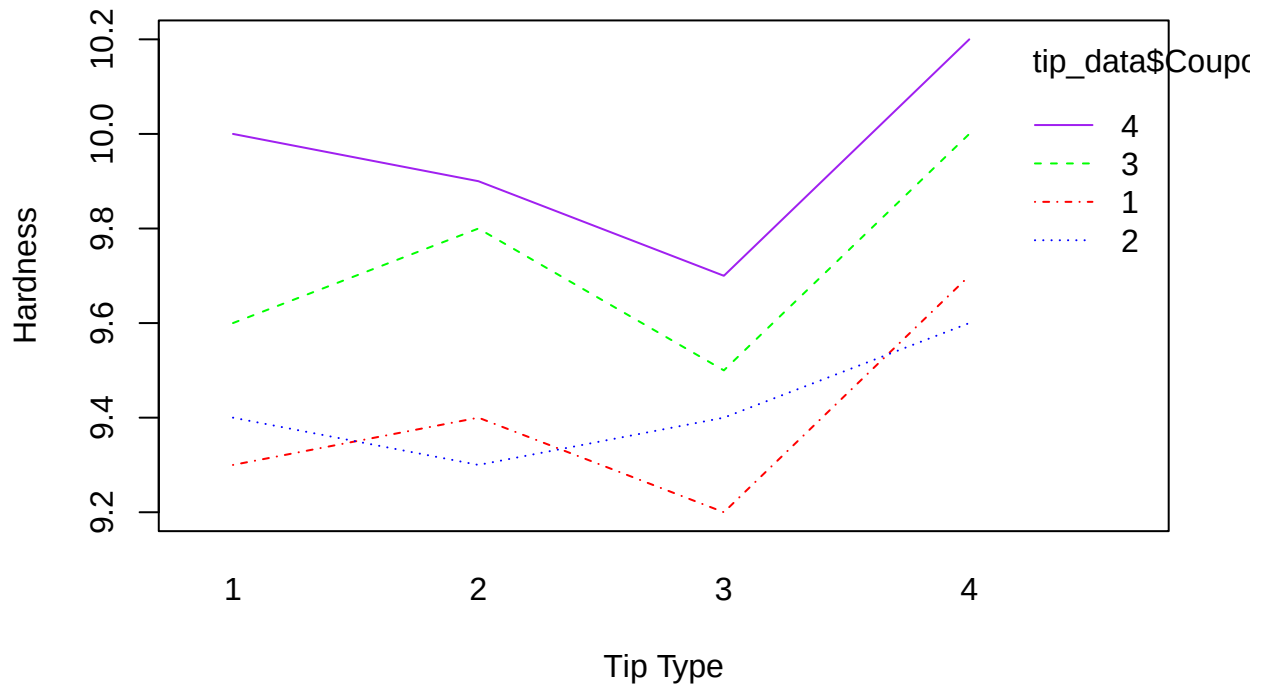
- p -value of Tip (0.00196) is lower than our $\alpha = 0.05$. Thus we reject null hypothesis H_0 and can conclude that at least one tip produces a different hardness.

5. Use an interaction plot to assess the additive assumption.

Code:

```
interaction.plot(
  x.factor = tip_data$Tip,
  trace.factor = tip_data$Coupon,
  response = tip_data$Hardness,
  col = c("red", "blue", "green", "purple"),
  main = "Tip vs. Coupon Interaction Plot",
  xlab = "Tip Type",
  ylab = "Hardness"
)
```

Tip vs. Coupon Interaction Plot



Answer:

- Some variation in slopes across Coupon, but no extreme crossing of Coupon lines. SD shows presence of minor interactions but additive assumption not violated.

In a study to evaluate the impact of a certain bacterial strain (mycoplasma) on the development of birds, birds inoculated with bacteria or a control injection were studied in both cold and warm temperature-controlled rooms. The response variable of interest was the bill length of birds in the rooms. The rooms served as the experimental units, with either cold or warm temperatures, and contained either control or mycoplasma-inoculated birds. Two runs of each combination were performed. The data is given below:

Bact	Warm	Cold
Control	40.37, 41.71	39.77, 40.23
Mycoplasma	40.21, 40.78	39.19, 38.95

6. Enter the data into R.

Code:

```
bill_data <- tibble(
  Bact = rep(c("Control", "Mycoplasma"), each = 4),
  Temp = rep(c("Warm", "Cold"), times = 4),
  Bill_Len = c(40.37, 39.77, 41.71, 40.23, 40.21, 39.19, 40.78, 38.95)
)

bill_data
```

```
## # A tibble: 8 x 3
##   Bact      Temp Bill_Len
##   <chr>    <chr>    <dbl>
## 1 Control  Warm      40.4
## 2 Control  Cold      39.8
## 3 Control  Warm      41.7
## 4 Control  Cold      40.2
## 5 Mycoplasma Warm      40.2
## 6 Mycoplasma Cold      39.2
## 7 Mycoplasma Warm      40.8
## 8 Mycoplasma Cold      39.0
```

- Construct an ANOVA table that includes the main effects for bacteria and temperature and their interaction. Test whether the main effects impact bill length.

Code:

```
# Convert to factor
bill_data <- bill_data %>%
  mutate(
    Bact = as.factor(Bact),
    Temp = as.factor(Temp)
  )

# Perform ANOVA
bill_anova <- aov(Bill_Len ~ Bact * Temp, data = bill_data)

# Summarize ANOVA results
anova_results <- summary(bill_anova)
anova_results
```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## Bact       1  1.0878   1.0878    3.642  0.1290
## Temp       1  3.0381   3.0381   10.171  0.0332 *
## Bact:Temp   1  0.0741   0.0741    0.248  0.6446
## Residuals   4  1.1949   0.2987
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

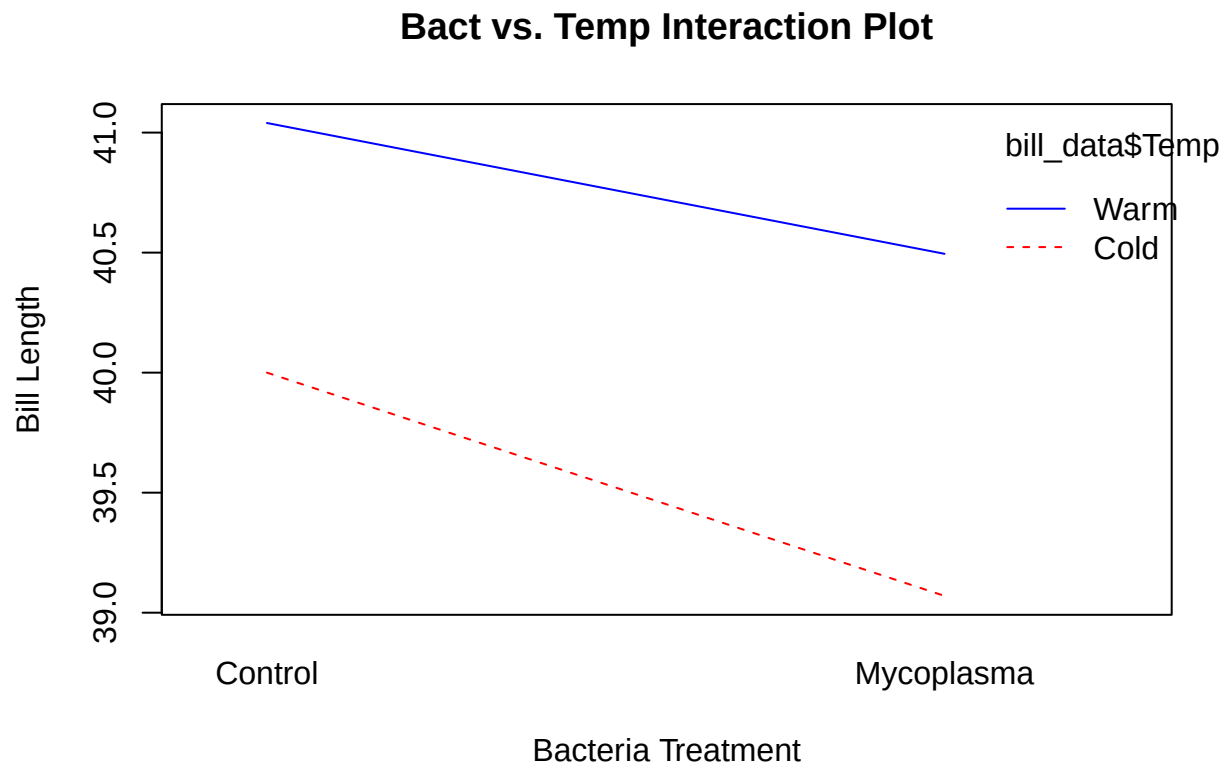
Answer:

- Bacteria: p-value (0.129) is greater than our alpha (0.05). We can conclude that bact treatment does not have effect on bill length.
 - Temperature: p-value (0.0332) is less than our alpha (0.05), indicating that temperature has statistically significant effect on bill length.
 - Bact:Temp (Interaction): p-value (0.64) greater than alpha (0.05), thus there is no significant interaction between bacteria treatment and temperature.
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8. Make an interaction plot and comment on the interaction effect.

Code:

```
interaction.plot(  
  x.factor = bill_data$Bact,  
  trace.factor = bill_data$Temp,  
  response = bill_data$Bill_Len,  
  col = c("red", "blue"),  
  main = "Bact vs. Temp Interaction Plot",  
  xlab = "Bacteria Treatment",  
  ylab = "Bill Length"  
)
```



Answer:

- Our lines for Warm and Cold do not cross, confirming our non-significant interaction shown by ANOVA (p-val: 0.64). Thus we can conclude that the effect of bacteria on bill length is independent of temperature.
- Result: Interaction between Bacteria and Temperature does not significantly affect Bill Length.